

SPEI3 Forecasting Scheme — v3

1. Inputs (Before Preprocessing)

Input group	Variable	Temporal coverage	Spatial resolution	Storage	Source
Climatological	Precipitation (mm/mon)	1901–2024	0.5° × 0.5°	NetCDF	CRU
	Min, Max temperature (°C)	1901–2024	0.5° × 0.5°	NetCDF	CRU
	PET (mm/day)	1901–2024	0.5° × 0.5°	NetCDF	CRU
	Solar radiation (J·m ⁻² / W·m ⁻²)	1940–2025	0.25° × 0.25°	NetCDF	ERA5
	Wind speed at 2 m (m·s ⁻¹)	1940–2025	0.25° × 0.25°	NetCDF	ERA5
	VPD (hPa)	1940–2025	0.25° × 0.25°	NetCDF	ERA5
	TCWV (kg/m ²)	1940–2025	0.25° × 0.25°	NetCDF	ERA5
	Rootzone Soil Moisture (RZSM, 0–100 cm, m³/m³)	1940–2025	0.25° × 0.25°	NetCDF	ERA5 (derived from swvl1–swvl3)
Large-scale climate indices	NAO Index	1950–2026	—	Text	NOAA CPC
	ENSO Niño 3.4	1870–2025	—	NetCDF	NOAA PSL
	Mediterranean Oscillation (MO)	1940–2025	—	NetCDF	ERA5-derived (Gibraltar – Lod SLP, standardized 1981–2010)

Potential additions deferred to a future iteration if v1 underperforms:

- AMO (Atlantic Multidecadal Oscillation) — documented linkage to North African precipitation
- AO (Arctic Oscillation) — partial NAO redundancy, possible additive skill

2. Preprocessing

- a. Align time period **1950–2024** for all input variables.
- b. Upscale ERA5 → 0.5° via conservative area-weighted averaging. All gridded data: 64 × 64 at 0.5°.

c. Missing-value handling (explicit):

- Temporal interpolation up to 2 consecutive missing months
- Drop cells with > 5 % missing in the training period of any fold

d. RZSM derivation:

- Source: ERA5 volumetric soil water layers 1–3 (swvl1: 0–7 cm; swvl2: 7–28 cm; swvl3: 28–100 cm)
- Depth-weighted average to 0–100 cm: $RZSM = (7 \cdot swvl1 + 21 \cdot swvl2 + 72 \cdot swvl3) / 100$
- Upscale to 0.5° via conservative area-weighted average (as for the other ERA5 variables)

e. Document SPEI3 computation method in study metadata: distribution (log-logistic), parameter estimator (L-moments), and fitting period. v1 default: full-record fit (1950–2024) with a sensitivity test on the last fold.

f. Stationarity diagnostics (run once, report in supplementary):

- Mann-Kendall trend test on annual mean SPEI3, per cell
- KS test on SPEI3 distributions for 1950–1989 vs 1990–2024
- Diagnostic only — does not gate modeling

3. Feature Engineering

3.1 Target Construction (Lead-Specific)

$y(t) = \text{SPEI3}(t + L)$, $L \in \{1, 3, 6\}$ months

Three independent datasets: **Dataset_L1**, **Dataset_L3**, **Dataset_L6**.

- Training: all months 1950–2024
- Evaluation: target month $\in$ [Nov–Feb] only

3.2 Predictor Set at Forecast Issue Time t

(A) Contemporary predictors (time t)

Precip, Tmax, Tmin, PET, VPD, Wind, Solar, TCWV, RZSM, ENSO, NAO, MO, SPEI3

(B) Lag selection methodology

- **PACF as primary tool** (not ACF), computed per CV fold on training data only.
- Test lags 1–12 uniformly for **all variables** — let the data prune.
- Significance threshold: **|PACF| > 0.20**, with a sensitivity test at 0.10 and 0.30 reported in appendix. This is much stricter than 95 % statistical significance (Bartlett's threshold = $1.96/\sqrt{N} \approx 0.065$ for $N = 900$, ≈ 0.113 for $N = 300$ winter-only); 0.20 is a deliberate practical-relevance filter rather than a significance filter.
- **CCF screening** between SPEI3 at time t and each candidate predictor at time $t - k$, restricted to **target months $t \in \{\text{Nov, Dec, Jan, Feb}\}$** since that is the evaluation season. Predictors whose CCF crosses $|CCF| > 0.20$ at any lag 1..12 are included at that lag.
- **Selected lags = union of PACF-passing lags and CCF-passing lags.** Variables with no surviving lags are dropped from the lagged feature set (the contemporary feature is unaffected).

Spatial aggregation for lag selection. PACF and CCF require a single 1-D time series per variable, but the inputs are gridded. The procedure:

1. Load the Morocco boundary from `shapefiles/MAR_adm0.shp` (164 cells inside, out of 4096 in the full grid).
2. Restrict the gridded variable to Morocco cells.
3. **Approach A (default)** — `spatial_mean` of the Morocco cells, then PACF + CCF on the resulting 1-D series.
4. **Approach B (alternative)** — PACF + CCF on each Morocco cell's 1-D series, then average those 164 spectra; threshold applied to the averaged spectrum. Implemented for transparency / sensitivity; reported in the appendix.

The two approaches agree closely for region-coherent signals (NAO, ENSO, MO, SPEI3, precip) and differ slightly for spatially-heterogeneous variables (RZSM, TCWV, VPD) where Approach A retains more long lags.

Important: spatial averaging is used *only* for lag selection. The CV / modeling pipeline (Sections 4–7) treats every (cell, time) sample as an independent observation, with (`lat`, `lon`) included as features (Section 3.2 D). No spatial averaging is applied at training or inference time.

- For linear models: hand all surviving lags to **Lasso / Elastic Net** for final pruning.
- For RF / XGBoost: use the PACF + CCF output directly (trees handle redundancy).

Variables explicitly tested for long-memory lags (up to 12): SPEI3, RZSM, ENSO, NAO, MO, TCWV, Precip, VPD. Fast-response variables tested for lag 1 only by default: Tmin, Tmax, Solar, Wind (extendable if PACF justifies).

(C) Seasonal encoding

- $\sin(2\pi \cdot m / 12)$, $\cos(2\pi \cdot m / 12)$ where m is month index.

(D) Spatial encoding

- `latitude`, `longitude` as features for the **global model** (Section 7.2).

4. Cross-Validation Strategy

5-fold rolling-origin, expanding training window, continuous test windows

Test windows are **contiguous** so the per-fold out-of-sample predictions stitch into an unbroken 2000–2024 array suitable for pooled-metric computation.

Fold	Train (planned)	Validate (planned, 8 y)	Test (5 y, continuous)
1	1950-01 → 1991-12 (42 y)	1992-01 → 1999-12	2000-01 → 2004-12
2	1950-01 → 1996-12 (47 y)	1997-01 → 2004-12	2005-01 → 2009-12
3	1950-01 → 2001-12 (52 y)	2002-01 → 2009-12	2010-01 → 2014-12
4	1950-01 → 2006-12 (57 y)	2007-01 → 2014-12	2015-01 → 2019-12
5	1950-01 → 2011-12 (62 y)	2012-01 → 2019-12	2020-01 → 2024-12

Pooled out-of-sample series: **2000-01 → 2024-12 unbroken (25 y; ≈ 100 winter target months $\times$ spatial-pooling cells).**

The "planned" sizes above are subsequently shrunk by a per-fold **boundary gap** (quarantine) — see §4.1.

Leakage controls:

- Target shifting performed before splitting; no future predictors.
- Per-fold standardization (Section 5).
- Per-fold feature selection (Section 6).
- Per-fold PACF / CCF lag selection (Section 3.2 B).
- **Per-fold boundary gap (quarantine) — Section 4.1.**

Aggregation across folds:

- **Pooled:** concatenate predictions across folds → compute headline metric once. With the continuous-test design this is exactly a single, unbroken 25-year out-of-sample array (100 winter months $\times$ cells). Pooled metrics are the primary reporting unit.
- **Per-fold:** same metrics reported per fold in supplementary table to show stability. Per-fold winter sample is small (20 months), so per-fold metrics are noisy by design; do not over-interpret single-fold differences.
- **Winter-only and all-months reporting:** the headline metric is computed on **winter target months only** ($t \in \{\text{Nov, Dec, Jan, Feb}\}$). A parallel **all-months evaluation** is computed on every test month and reported alongside as a supplementary diagnostic — it uses 4 $\times$ more evaluation samples and tests whether skill is uniform across seasons or concentrated in the winter target.

4.1 Boundary gap (quarantine)

Two distinct leakage paths must be closed at every train→val and val→test boundary. The quarantine width must dominate **both** simultaneously.

Path 1 — Lagged-predictor leakage. A val (or test) sample at feature time t uses lagged predictors $\text{SPEI3}(t-1)$, $\text{SPEI3}(t-2)$, ..., $\text{SPEI3}(t-K)$, where K is the deepest selected lag for the fold. If those $t-k$ months are themselves training **targets**, the model has effectively fit (X, y) at the same months whose feature values it now reads. Information flows backward through the lag features.

Path 2 — Rolling-window contamination. SPEI3 is a 3-month standardized index: $\text{SPEI3}(\tau)$ is a deterministic function of precipitation at $\{\tau-2, \tau-1, \tau\}$. The same *physical precipitation month* can therefore appear inside **both** a training target $\text{SPEI3}(t_{\text{train}} + L)$ **and** a val/test predictor $\text{SPEI3}_{\text{lag}}(k)(t')$, even when those targets and predictors are at different calendar months and $t' \neq t_{\text{train}} + L$. The naïve "lead + 2" purge — which covers only the rolling window of the contemporaneous feature — misses this whenever $K > 0$.

Derivation of the required gap. Only variables whose lag features share **raw precipitation or PET ingredients** with the SPEI3 target can leak through shared months. We call these the *precip-touching* variables. In our feature set: SPEI3 (and its lags), precip (and its lags), and PET (only contemporaneous in our config). All other variables — NAO, ENSO, MO, RZSM, TCWV, VPD, wind, solar, tmin, tmax — are derived from **independent data sources** (NOAA pressure indices, ERA5 atmosphere, ERA5 soil moisture), so their lag features have only legitimate statistical dependence with future SPEI3; they impose **no** quarantine constraint.

Per precip-touching feature class, the precip-month footprint at val feature time t is:

Feature class	Precip months touched	Disjoint-from-train-target constraint
Contemporaneous precip(t), PET(t)	$\{t\}$	$\text{gap} \geq L$
Lagged precip_lag(k)(t), PET_lag(k)(t)	$\{t - k\}$	$\text{gap} \geq L + k$
Contemporaneous SPEI3(t)	$[t - 2, t]$	$\text{gap} \geq L + 2$
Lagged SPEI3_lag(k)(t)	$[t - k - 2, t - k]$	$\text{gap} \geq L + k + 2$

Taking the maximum over all present precip-touching feature classes, define the **effective lag**:

$$K_{\text{eff}} = \max(0, \max_{\{k \in \text{SPEI3 lags}\}} k, \max_{\{k \in \text{precip lags}\}} (k - 2), \max_{\{k \in \text{PET lags}\}} (k - 2))$$

so that

$$\text{gap} = L + K_{\text{eff}} + 2$$

closes all strict precip-month leakage. The floor at 0 enforces $\text{gap} \geq L + 2$, which is the constraint from the contemporaneous SPEI3 feature (always present).

Adaptive K_{eff} . K_{eff} is determined per fold from PACF on the autoregressive variables + winter-only CCF on the climate indices, run over the **provisional** train slice; the per-variable lag dict is then mapped through the `compute_quarantine_max_lag` helper (precip-touching variables only) to obtain K_{eff} . The gap $L + K_{\text{eff}} + 2$ is applied and indices are finalized:

```
for fold in cv_folds:
    1. Provisional split using planned windows above.
    2. Run PACF + winter-only CCF on the provisional train → selected_lags
       (a dict variable_name → list of selected lag depths).
    3. Compute  $K_{\text{eff}}$ :
         $K_{\text{eff}} = \max($ 
            0,
             $\max(\text{selected\_lags}["\text{spei3}"], \text{default}=0),$ 
             $\max(\text{selected\_lags}["\text{precip}"], \text{default}=2) - 2,$ 
             $\max(\text{selected\_lags}["\text{pet}], \text{default}=2) - 2,$ 
        )
        (variables not in the precip-touching set are ignored).
    4. Refine indices:
        gap = L +  $K_{\text{eff}}$  + 2
        train_indices = months in [train_start .. val_start - gap - 1]
        val_indices   = months in [val_start   .. test_start - gap - 1]
        test_indices  = months in [test_start  .. test_end]      # never
    shrunk
    5. Fit fold-wise standardizer on train_indices only.
    6. Train; tune HPs on val_indices; predict on test_indices.
```

Test windows are **never** shrunk — the pooled out-of-sample stitching covers the continuous 2000-01 → 2024-12 span regardless of lead.

Effective fold sizes for the worst case $L = 3$, $K_{\text{eff}} = 12 \rightarrow \text{gap} = 17$ (e.g. SPEI3 PACF selects lag 12). For lighter precip-touching lag selections the gap shrinks toward the $L + 2$ floor; the per-fold log emitted by the pipeline records both K_{eff} and the resulting train/val sizes.

Fold	Effective train	Effective val (HP tuning)	Test (5 y, unchanged)
1	1950-01 → 1990-07 (40 y 7 mo)	1992-01 → 1998-07 (6 y 7 mo)	2000-01 → 2004-12
2	1950-01 → 1995-07 (45 y 7 mo)	1997-01 → 2003-07 (6 y 7 mo)	2005-01 → 2009-12
3	1950-01 → 2000-07 (50 y 7 mo)	2002-01 → 2008-07 (6 y 7 mo)	2010-01 → 2014-12
4	1950-01 → 2005-07 (55 y 7 mo)	2007-01 → 2013-07 (6 y 7 mo)	2015-01 → 2019-12
5	1950-01 → 2010-07 (60 y 7 mo)	2012-01 → 2018-07 (6 y 7 mo)	2020-01 → 2024-12

For $L = 6$ each train/val end shifts back another 3 months; for $L = 1$ each shifts forward 2 months relative to the table above. The cost (5 extra quarantined months per boundary vs the previous K -only gap) is small relative to the ~500+ months of train data per fold.

5. Standardization

Fold-wise standardization using **training-period statistics only**. The (μ, σ) statistics are always computed from the training slice exclusively (no leakage from val or test), and are always restricted to **Morocco cells** via the `MAR_adm0.shp` shapefile mask (164 of 4096 grid cells), so that non-Morocco grid points (e.g. Saharan fringe with extreme aridity distortion) don't bias the regional statistics.

Pre-standardized exception list — do NOT re-standardize:

- ENSO, NAO, MO (and their lags)
- **SPEI3 (and its lags)** when used as a predictor
- `target` (SPEI3 shifted forward by the lead)

All other variables — **including RZSM** — are standardized fold-wise.

Two variants of (μ, σ) computation are implemented, and the choice is tied to the modeling-unit choice in §7.2:

Variant	(μ, σ) scope	Spatial heterogeneity	Used by
<code>FoldStandardizer</code> (pooled, default)	One (μ, σ) per variable, pooled across (time × Morocco cells).	Preserved in z-space — Marrakech and Tangier sit at different z-values reflecting their position in the regional distribution. The global model uses <code>(lat_feat, lon_feat)</code> to learn cell-specific behavior.	Global model (v1 headline; §7.2 default)

Variant	(μ, σ) scope	Spatial heterogeneity	Used by
PerCellStandardizer (per-cell)	One (μ, σ) per (variable, lat, lon) cell, computed from that cell's own training-period time series.	Removed — every Morocco cell normalized to $(\mu=0, \sigma=1)$ over its own history. (lat, lon) features should be dropped (they become zero-variance constants within a cell).	Per-cell sensitivity test (§7.2)

Both variants compute statistics from training data only, apply the Morocco mask, and respect the same exception list. Non-Morocco cells in the **PerCellStandardizer** pass through unchanged (identity transform).

Procedure per fold:

1. Compute mean / std on training data only (variant-dependent scope).
2. Apply the transform to training, validation, and test sets.
3. Exception-list variables pass through untouched, both variants.

The **03_cv_visualization** notebook demonstrates a side-by-side comparison of both variants on a representative fold, including verification that the exception list bypasses both standardizers.

6. Feature Selection (per fold)

A single selection method is used: **Lasso / Elastic Net** for linear models (selection happens during fitting via the L1 penalty). Tree-based models use no explicit selection — they tolerate redundant features natively. Permutation importance (RF) and SHAP (XGBoost) are computed for **interpretability and reporting only**, not used to drop features.

Method	Role	Applied to
Lasso / Elastic Net	Selection (during fit via L1)	Linear models — primary
Permutation importance	Diagnostic (post-hoc ranking)	Random Forest
SHAP values	Diagnostic (post-hoc ranking + interpretability)	XGBoost

Pipeline per model family:

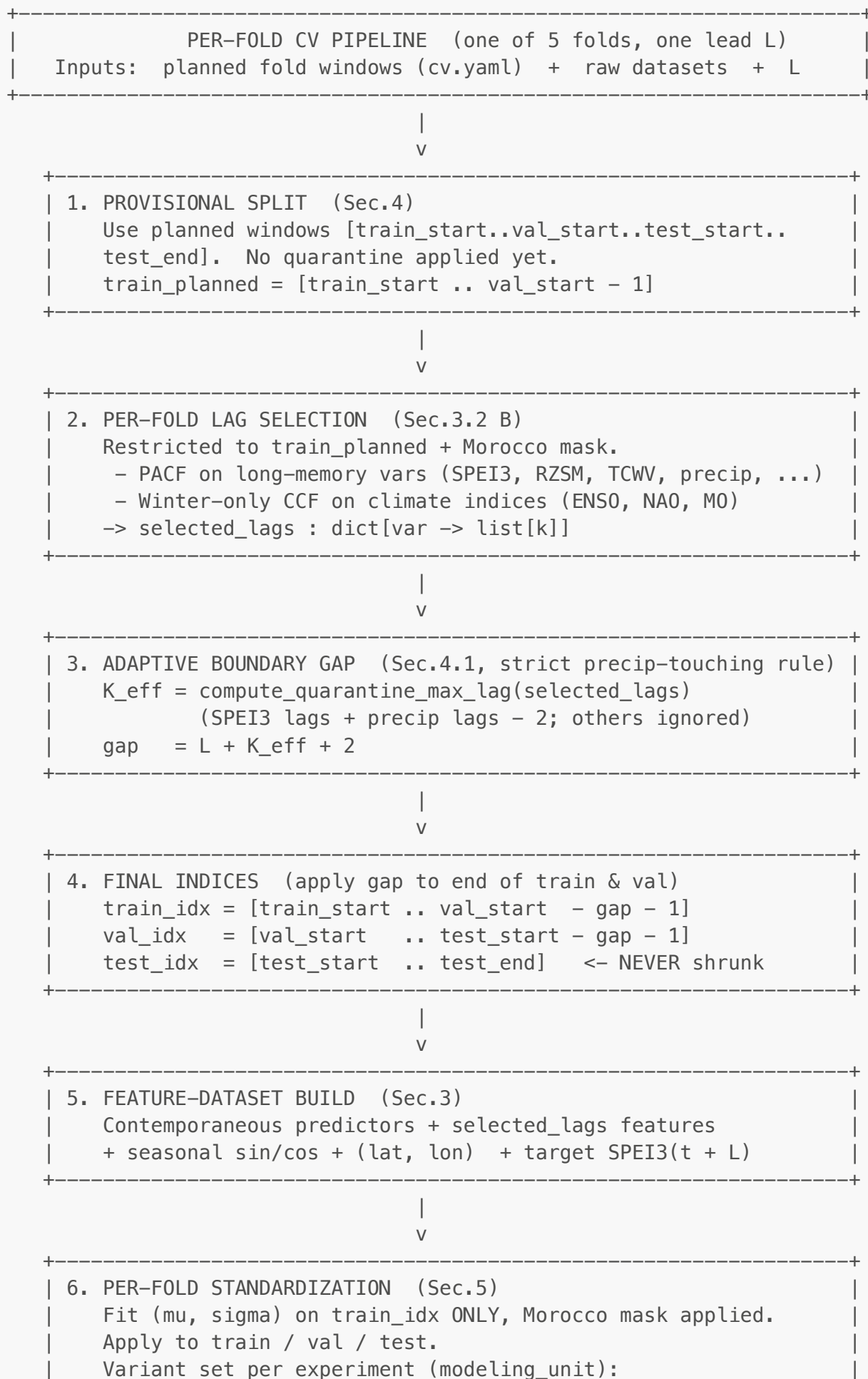
- **Linear models** (OLS / Ridge / Lasso / Elastic Net): All Section-3 features pass to the fitter; Lasso / Elastic Net's L1 penalty produces the final sparse coefficient vector. Multicollinearity is absorbed by the L1 / L2 regularizer; no separate VIF pre-filter is applied.
- **RF / XGBoost**: All Section-3 features pass to the fitter. After training, permutation importance (RF) and SHAP (XGBoost) are reported as paper diagnostics.

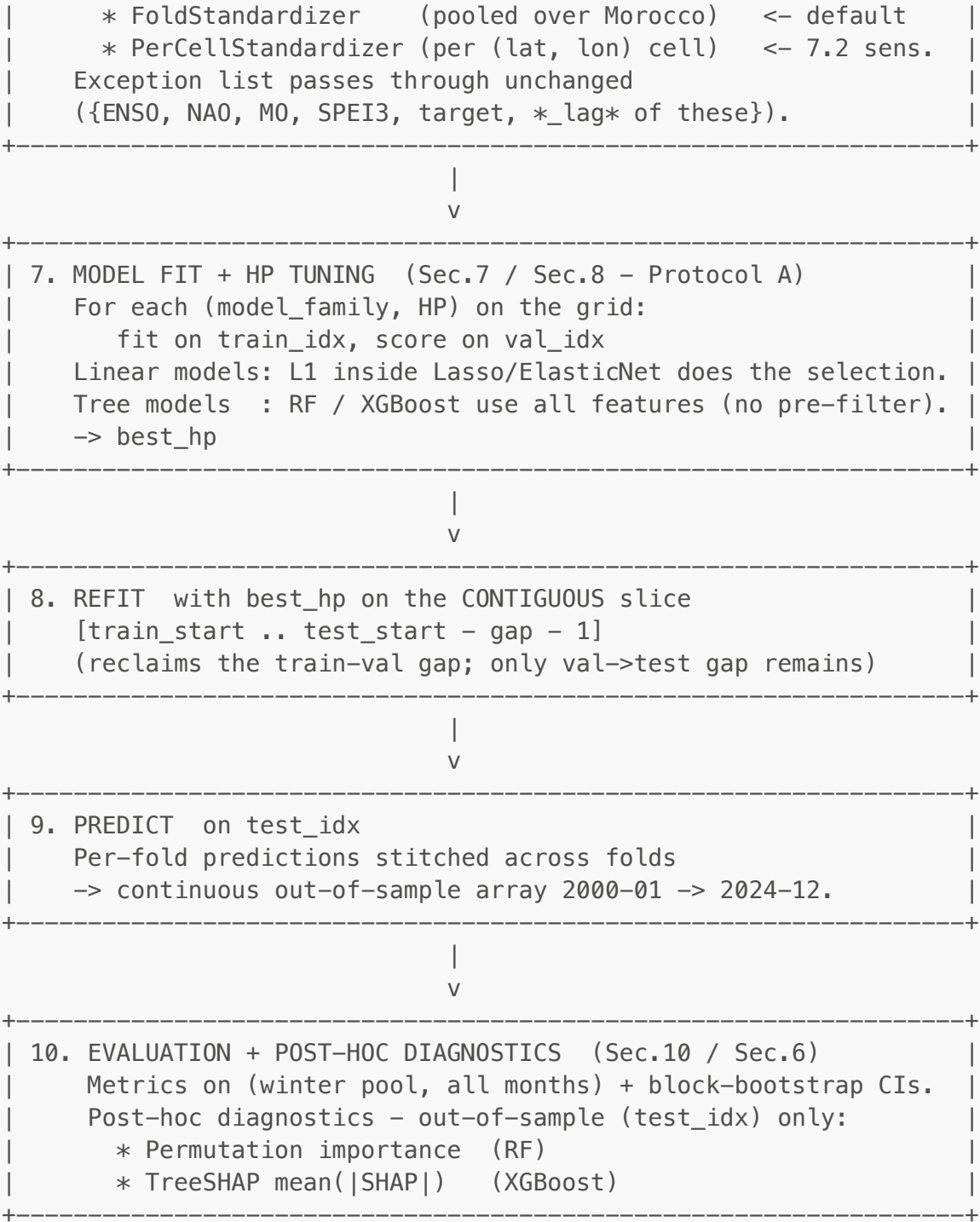
Selection is repeated per fold (Lasso is refit per fold), preserving the leakage discipline.

6.1 Per-fold pipeline at a glance

The diagram below traces the order of operations for one fold, tying together §3 (features), §4 (CV + boundary gap), §5 (standardization), §6 (selection), §7 (model fit), §8 (HP tuning), and §10 (evaluation).

Every step uses **training-data-only** statistics; the test slice is never touched until step 9.





Steps 1–6 are pure per-fold preprocessing; steps 7–10 are per (fold, model_family, lead). Each fold is independent — the orchestrator (Phase 10) parallelises the outer loop trivially.

7. Model Configurations

7.0 Baselines

Baseline	Forecast formula	Purpose
Climatology	per-cell per-calendar-month training mean	Reference for skill scores
Persistence	$SPEI3(t+L) = SPEI3(t)$	Inertia benchmark

Baseline	Forecast formula	Purpose
AR(p)	Linear regression on lagged SPEI3 only	Univariate autoregressive benchmark (subsumes damped persistence at $p = 1$)

7.1 ML Models

Model	Input	Purpose
Linear (OLS)	Tabular	Statistical baseline
Ridge	Tabular	L2-regularized linear
Lasso	Tabular	L1-regularized (also feature selector)
Elastic Net	Tabular	L1 + L2 regularized
Random Forest	Tabular	Nonlinear ensemble
XGBoost	Tabular	Gradient boosting (primary tabular benchmark)
LSTM	Lagged sequences	Temporal memory (deferred)
CNN	Spatial grids	Spatial pattern extraction (deferred)
CNN-LSTM / ConvLSTM	Spatio-temporal sequences	Joint spatio-temporal (deferred)

7.2 Modeling Unit

- **Default:** global model with (lat, lon) as features — one model per (model family, fold), trained on all 64×64 cells jointly.
- **Sensitivity test:** per-cell models for the best-performing family.

8. Hyperparameter Tuning

Protocol A — tune on validation, refit on train + val, evaluate on test:

```
for fold in cv_folds:
    best_hp = search(model_type, X_train, y_train, X_val, y_val)
    final_model = model_type(best_hp).fit([train_start, test_start - gap - 1], y_train U y_val)
    preds = final_model.predict(X_test)
```

- **Linear models** (Ridge, Lasso, Elastic Net) — **grid search**. Their spaces are small (13–35 combos) and exhaustive enumeration is reproducible and cheap.
- **Tree models** (RF, XGBoost) — **Optuna TPE** with `n_trials=40` per (fold, lead). The categorical search spaces are 72 and 144 respectively; sampling at `n_trials=40` cuts tree wall time by ~3x compared with exhaustive grid while landing on near-best HPs (TPE focuses sampling on high-skill regions after a brief uniform-exploration phase). Grid is still available for either backend.

- **RF $n_{\text{estimators}}$ trimmed to {200, 500}** (dropping 1000): the $n=1000$ configs dominate per-trial cost without meaningful gain in best-HP quality on Morocco-scale data (5 folds $\times$ ~80k pooled samples per fit). The trim cuts RF's average per-trial cost ~30–40 %; combined with Optuna's **$n_{\text{trials}}=40$** budget, this reduces total tree wall time from ~60–80 h (full grid) to ~9 h on a typical laptop.

Search spaces:

Model	Space	Backend	Sampled
Ridge / Lasso	$\alpha \in \text{logspace}(-3, 3, 13)$	grid	all 13
Elastic Net	$\alpha \in \text{logspace}(-3, 3, 7)$; $l1_ratio \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$	grid	all 35
RF	$n_{\text{estimators}} \in \{200, 500\}$; $\text{max_depth} \in \{\text{None}, 5, 10, 20\}$; $\text{min_samples_leaf} \in \{1, 5, 20\}$; $\text{max_features} \in \{\text{sqrt}, 0.5, 1.0\}$	Optuna TPE	40 of 72
XGBoost	$\text{max_depth} \in \{4, 6, 8\}$; $\text{lr} \in \{0.05, 0.1\}$; $\text{subsample} \in \{0.7, 1.0\}$; $\text{colsample_bytree} \in \{0.7, 1.0\}$; $\text{reg_lambda} \in \{0.1, 1.0, 10.0\}$; $\text{min_child_weight} \in \{1, 5\}$; $n_{\text{estimators}}$ via early stopping on val	Optuna TPE	40 of 144

9. Forecast Generation

For each issue month t :

1. Extract predictors up to month t .
2. Apply trained model for lead L (one model per (family, fold, lead)).
3. Produce forecast $\text{SPEI3}(t + L)$.
4. Retain forecasts verifying in Nov–Feb only.

10. Evaluation Metrics

10.1 Deterministic (continuous SPEI3) — headline

Metric	Formula	Role
MAE	$\text{mean}(\hat{y} - y)$	Headline — error magnitude
RMSE	$\sqrt{[\text{mean}((\hat{y} - y)^2)]}$	Headline — error magnitude
Pearson r	$\text{corr}(\hat{y}, y)$	Headline — pattern correlation
ACC	$\text{corr}(\hat{y} - \text{clim}, y - \text{clim})$	Headline — anomaly correlation
MSSS vs climatology	$1 - \text{MSE}(\text{model}) / \text{MSE}(\text{climatology})$	Headline — % improvement over climatology

Metric	Formula	Role
MSSS vs persistence	$1 - \text{MSE}(\text{model}) / \text{MSE}(\text{persistence})$	Headline — added value beyond inertia

10.2 Optional metrics (implemented in code, not reported by default)

- **HSS at SPEI3 < -1.0** — binary categorical (drought / no drought). Available via the metrics config but not part of the v1 headline.
- **POD, FAR, CSI, ETS, multi-class HSS** — not implemented in v1; deferred to a follow-up if categorical evaluation becomes a focus.
- **Probabilistic (CRPS, Brier, reliability)** — deferred; requires quantile / probabilistic models.

10.3 Headline Metrics (committed)

The six deterministic metrics in §10.1 form the headline set. They are reported per (model, lead) on the **pooled** out-of-sample array (2000–2024 winter target months for the headline; all months for the supplementary table).

10.4 Reporting Structure

- Per lead time ($L = 1, 3, 6$).
- Pooled across folds (headline). Per-fold supplementary table is a planned addition — currently the pipeline emits one `fold='pooled'` row per (model, lead, metric, evaluation_window).
- Spatial: per-cell skill maps + pooled metrics across cells.
- **Winter-only** evaluation ($t \in \text{Nov–Feb}$) is the **headline** unit; **all-months** evaluation is reported as a supplementary diagnostic alongside (uses 4× more samples, lets us check whether skill is winter-specific or season-uniform).
- **Block bootstrap 95 % CIs** on all headline metrics:
 - Stationary bootstrap; mean block ≈ 12 months.
 - For winter-only metrics: year-blocks (full Nov–Feb season per block).
 - 1000 replicates.

11. Outputs

The pipeline (`ExperimentRunner` in `droughtmodel/pipeline.py`) produces a fixed set of artefacts per experiment. Output filenames are prefixed by `exp.file_prefix` (default `""`) so smoke and production runs can coexist without collision. The paths below assume the default prefix.

Pipeline artefacts (written by `scripts/03_run_experiment.py`):

File	Description
<code>results/predictions/pooled_allLeads.nc</code>	NetCDF with one variable per model (<code>pred_{name}</code>) plus <code>truth</code> , all of shape (<code>lead, time, lat, lon</code>); the <code>time</code> axis is the continuous 2000–2024 OOS stitch shared across leads

File	Description
results/metrics/metrics_allLeads.csv	Pooled headline metrics with block-bootstrap CIs; one row per (model, lead, metric, evaluation_window); fold='pooled' (see §10.4)
results/logs/fold_runs.csv	One row per (fold, lead, model): best HP, best val score, search duration, fit duration, K_eff , boundary_gap , n_trials , n_features_total , n_features_retained
results/logs/feature_status.csv	One row per (fold, lead, model, feature): importance , retained (bool), kind (coef / gini / gain)

Phase 11 presentation-layer artefacts (built by the results notebooks under [notebooks/](#)):

Output	Notebook	Description
Headline skill tables (CSV + LaTeX)	09_paper_figures.ipynb	Pooled metrics with CIs + significance markers, ready for direct paper inclusion
Per-cell ACC and MSSS maps	07_spatial_skill_maps.ipynb	Model × lead grids; latitudinal skill profile by quartile band
Difference maps (ML – baseline)	07 , 09	Where each ML model adds value over the strongest baseline
L1 retention heatmaps	08_feature_importance.ipynb	ElasticNet / Lasso retention across folds per (lead, feature)
Top-feature bar charts	08 , 09	Built-in tree importance (Gini / gain) — see §11 note below on permutation / SHAP
Lift-over-best-baseline table	06_results_tabular_ml.ipynb	Per-(model, lead, metric) lift over the strongest baseline
Lag-selection diagnostics	02_feature_engineering_diagnostics.ipynb	PACF / CCF plots per variable per fold
Stationarity diagnostics	01_eda.ipynb	Mann-Kendall and KS-test results per cell

Note on feature importance. The current `feature_status.csv` carries the **built-in** importance for each model — standardized coefficients for linear, Gini for RF, gain for XGBoost. Built-in tree importances under-credit correlated features (see §6 notes). Production-quality permutation importance (RF) and TreeSHAP (XGBoost) require the fitted model object, which the pipeline does not currently persist. The helpers exist in `droughtmodel/selection.py`; adding model serialization + a post-hoc importance pass is a planned polish.

Summary of Experimental Structure (v3)

Dimension	Specification
Time period	1950–2024
Temporal resolution	Monthly
Spatial resolution	64 × 64 grid at 0.5° over Morocco
Target	SPEI-3 at L = 1, 3, 6 months
Evaluation season	Winter (Nov–Feb)
Cross-validation	5-fold rolling-origin, expanding train window, continuous test windows (2000–2024 unbroken)
Leakage control	Fold-wise standardization, feature selection, lag selection; strict target shifting; boundary-gap quarantine <code>gap = L + K_eff + 2</code> (precip-touching variables only) at train → val and val → test boundaries
Pre-standardized exceptions	ENSO, NAO, MO, SPEI3 (as predictor) , target
Climate drivers	ENSO, NAO, MO (+ optional AMO, AO in v2)
Lag selection	PACF + winter-only CCF on Morocco-masked spatial mean; threshold $ \cdot > 0.20$; sensitivity $\in \{0.10, 0.30\}$; Lasso finalizes for linear
Region mask for lag selection	<code>shapefiles/MAR_adm0.shp</code> (164 cells inside) — Approach A (spatial mean) default; Approach B (per-cell then mean) reported as appendix sensitivity
Modeling unit	Global model with (lat, lon) features (v1); per-cell as sensitivity
Standardization variants	<code>FoldStandardizer</code> (pooled over Morocco, default for global model); <code>PerCellStandardizer</code> (per (lat, lon), for §7.2 sensitivity)
Baselines	Climatology, persistence, AR(p)
Feature selection	Lasso / Elastic Net for linear (selection during fit); no explicit selection for trees; SHAP / permutation importance reported as diagnostics
ML models	OLS / Ridge / Lasso / Elastic Net, RF, XGBoost
DL models	LSTM, CNN, CNN-LSTM (or ConvLSTM) — deferred
HP tuning	Protocol A — tune on val, refit on train + val, eval on test

Dimension	Specification
HP search	Linear models: grid (Ridge/Lasso 13, ElasticNet 35). RF: Optuna TPE, n_trials=40 over a 72-combo categorical space (n_estimators trimmed to {200, 500}). XGBoost: Optuna TPE, n_trials=40 over a 144-combo categorical space; n_estimators via early stopping on val
Headline metrics (all deterministic)	MAE, RMSE, Pearson r , ACC, MSSS-vs-clim, MSSS-vs-persistence
Optional metric	HSS at SPEI3 < -1.0 (available in code, not in v1 paper headline)
Evaluation window	Winter-only (Nov–Feb) headline; all-months supplementary diagnostic
Uncertainty	Stationary block bootstrap, 1000 replicates, 95 % CI
Outputs	Forecasts, spatial skill maps, time series, importance, skill tables, CIs